

ETF5231 Group Assignment 4

Group7 GenieUS

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Question 1

Update your data from the ABS website and plot it. Do you see a COVID-19 effect? (3 marks)

```
extended_dat <- read_excel("GA4_data.xlsx", sheet = "Data1", skip = 9)

print_retail_act <- extended_dat |>
  rename(Month = "Series ID", y = "A3349773T") |>
  select(Month, y) |>
  mutate(Month = yearmonth(Month)) |>
  as_tsibble(index = Month)
```

```
plot_whole_series <- print_retail_act |>
  autoplot(y) +
  annotate(
    "rect",
    xmin = yearmonth("2020 Mar"),
    xmax = yearmonth("2022 Sep"),
    ymin = -Inf,
    ymax = Inf,
    alpha = 0.2,
    fill = "grey30"
  ) +
  labs(
    x = "Month",
    y = "Turnover (Millions of Dollars)"
  ) +
  theme_minimal()
```

```

plot_zoomed <- print_retail_act |>
  filter(Month >= yearmonth("2018 Jan")) |>
  autoplot(y) +
  annotate(
    "rect",
    xmin = yearmonth("2020 Mar"),
    xmax = yearmonth("2022 Sep"),
    ymin = -Inf,
    ymax = Inf,
    alpha = 0.2,
    fill = "grey30"
  ) +
  labs(
    x = "Month",
    y = "Turnover (Millions of Dollars)"
  ) +
  theme_minimal()

(plot_whole_series / plot_zoomed) +
  plot_annotation(
    title = "Turnover Series with COVID-19 Period Highlighted",
    subtitle = "Newspaper and book retailing, Australian Capital Territory"
  )

```

Turnover Series with COVID-19 Period Highlighted
Newspaper and book retailing, Australian Capital Territory

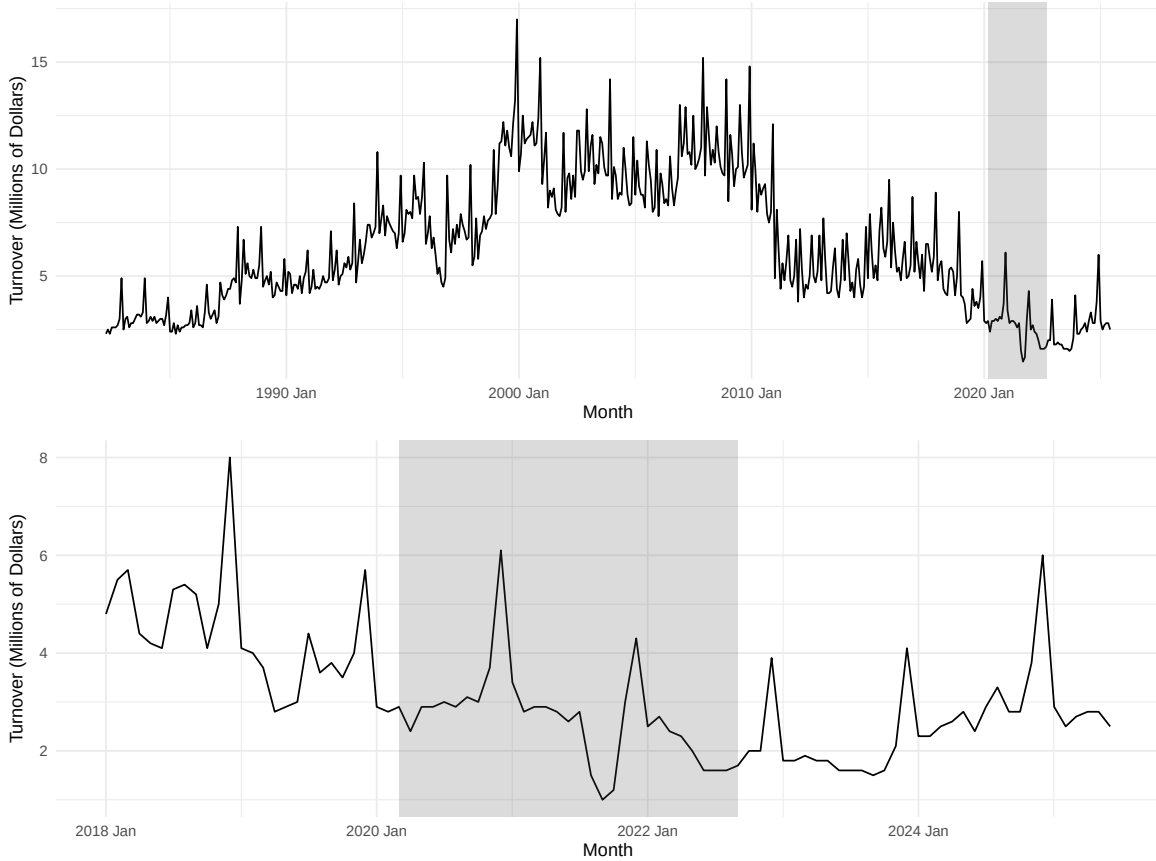


Figure 1: Updated turnover series from Apr 1982 to Jun 2025. The shaded area highlights March 2020 to Sep 2022, when ABS did not publish trend estimates due to COVID-related disruption and volatility. The left plot shows the whole series, while the right plot zooms in from Jan 2018 to show recent movements.

Figure 1 shows a long-term decline in turnover before COVID-19. During the COVID-disruption period, turnover remained low and volatile, with some sharp temporary movements, including the lowest point in the series. After the COVID-disruption period, turnover shows a slight recovery, but remains at a relatively low level, with the seasonal pattern still visible.

```
print_retail_act |>
  filter(Month >= yearmonth("2018 Jan")) |>
  gg_season(y, labels = "both") +
  labs(
    title = "Turnover (Jan 2018 - Jun 2025)",
```

```

subtitle = "Newspaper and book retailing, Australian Capital Territory",
x = "Month",
y = "Turnover (Millions of Dollars)"
) +
theme_minimal()

```

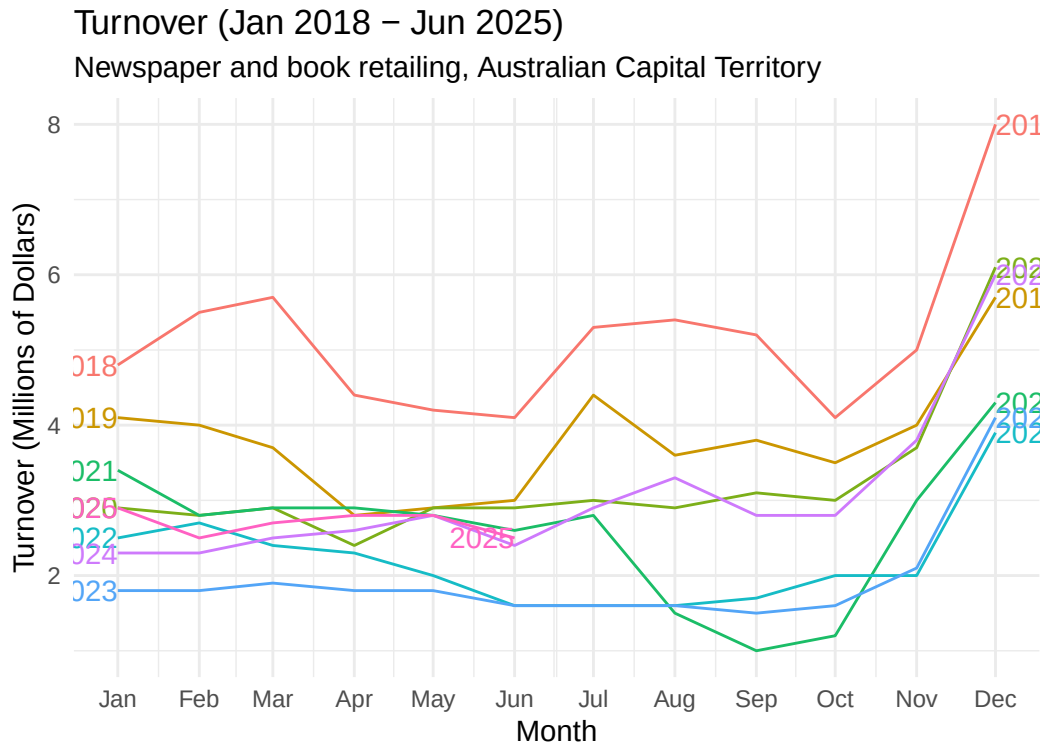


Figure 2

Figure 2 shows that turnover generally declined over time while retaining a consistent seasonal pattern, with higher values toward the end of the year, especially in December. During the COVID-disruption period, the 2021 line falls sharply from July to September and reaches the lowest point, suggesting a strong temporary weakening. After the COVID-disruption period, turnover increases slightly in some months, but remains relatively low compared with 2018–2020.

Overall, COVID-19 appears to have contributed to additional volatility and a temporary weakening around 2021, although the series was already following a long-term downward trend before the disruption period.

Question 2

Identify the best possible ARIMA model for your data and generate forecasts. Present a complete analysis and justifications for each step you take. There is no word limit here. No more hints will be provided - we leave it up to you to provide a full analysis. Treat this as a consulting job for a client wanting some ARIMA forecasts (20 marks)

Transformation

```
lambda_tbl <- print_retail_act |>
  features(y, guerrero)

lambda <- lambda_tbl$lambda_guerrero

lambda_tbl |>
  mutate(lambda_guerrero) |>
  rename(`Guerrero lambda` = lambda_guerrero) |>
  kable(digits = 3)
```

Table 1: Guerrero lambda for the updated series.

Guerrero lambda
0.424

```
p1 <- print_retail_act |>
  autoplot(y) +
  labs(title = "Original", x = "Month", y = "Turnover")

p2 <- print_retail_act |>
  autoplot(sqrt(y)) +
  labs(title = "Square-root", x = "Month", y = "sqrt(Turnover)")

p3 <- print_retail_act |>
  autoplot(log(y)) +
  labs(title = "Log", x = "Month", y = "log(Turnover)")

p4 <- print_retail_act |>
  autoplot(box_cox(y, lambda)) +
  labs(title = "Box-Cox using Guerrero lambda",
```

```

    x = "Month", y = "Box-Cox(Turnover)")

(p1 / p2 / p3 / p4) +
  plot_annotation(
    title = "Transformation comparison for ACT newspaper and book retail turnover",
    theme = theme(
      plot.title = element_text(size = 18, face = "bold", hjust = 0.5)
    )
  )

```

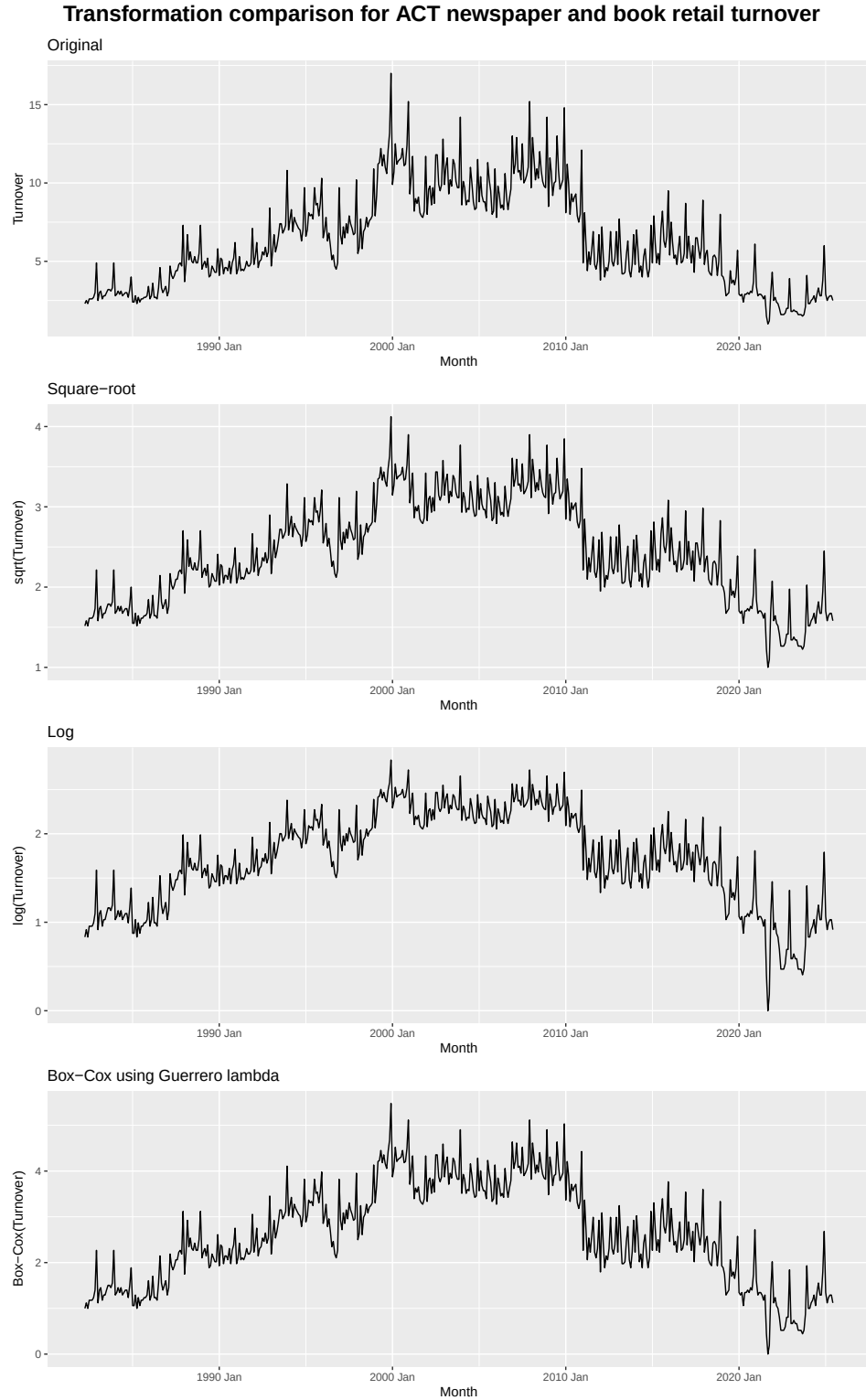


Figure 3: Comparison of transformations for the updated turnover series.

Figure 3 compares the original, square-root, log and Box-Cox transformations. The original series has changing variation over time, so a transformation is needed to stabilise the variance before ARIMA modelling. Guerrero's lambda is around 0.424, meaning log is too strong and the original scale is not stable enough. Square-root is a reasonable approximation, but Box-Cox uses the estimated lambda directly, so I use `box_cox(y, lambda)`.

```
print_retail_act |>
  autoplot(box_cox(y, lambda)) +
  labs(title = "Box-Cox transformed turnover series",
       x = "Month",
       y = "Box-Cox(Turnover)")
```

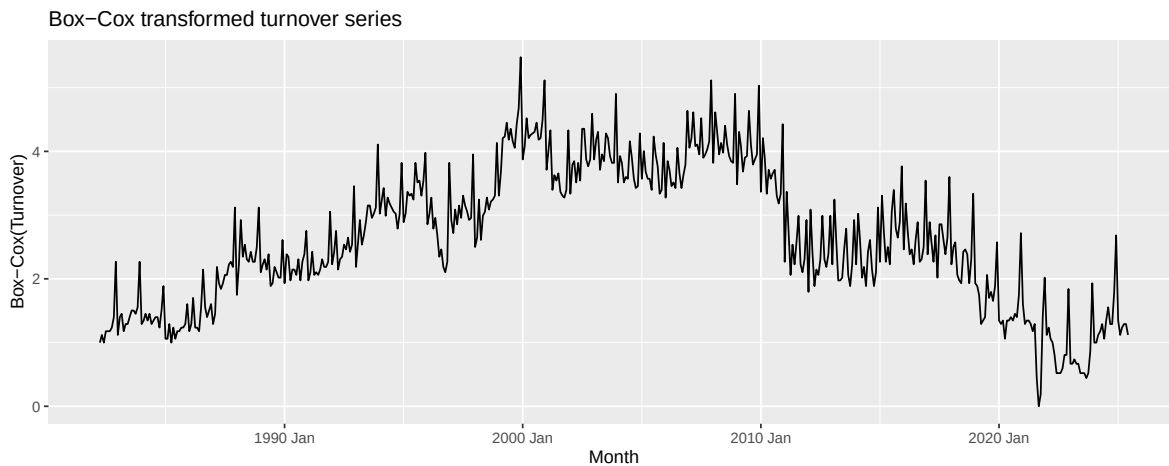


Figure 4: Box-Cox transformed turnover series.

Differencing and stationarity

Figure 4 still shows a long-run downward movement and regular monthly seasonality. Therefore, differencing is needed before selecting ARIMA orders.

```
print_retail_act |>
  autoplot(difference(box_cox(y, lambda), 12)) +
  labs(title = "Seasonally differenced Box-Cox transformed turnover",
       x = "Month",
       y = "Seasonal difference of Box-Cox(Turnover)")
```

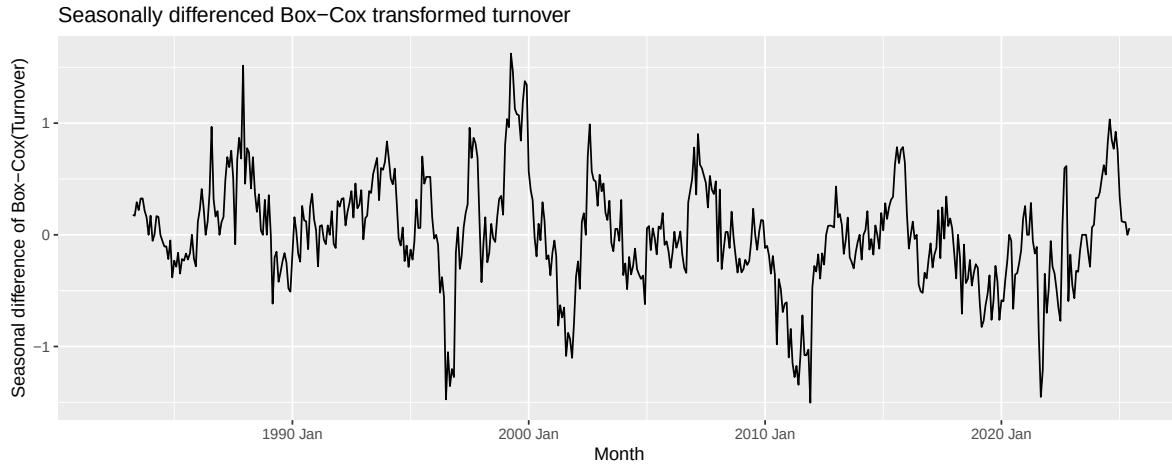



Figure 5: Seasonally differenced Box-Cox transformed turnover series.

Figure 5 shows that seasonal differencing removes much of the repeated monthly pattern, but the series still has long swings above and below zero. This suggests that the transformed series may not yet be fully stationary, so I check whether an additional first difference is needed.

```
print_retail_act |>
  autoplot(difference(box_cox(y, lambda), 12) |> difference()) +
  labs(title = "Seasonally and first differenced Box-Cox transformed turnover",
        x = "Month",
        y = "Differenced Box-Cox(Turnover)")
```

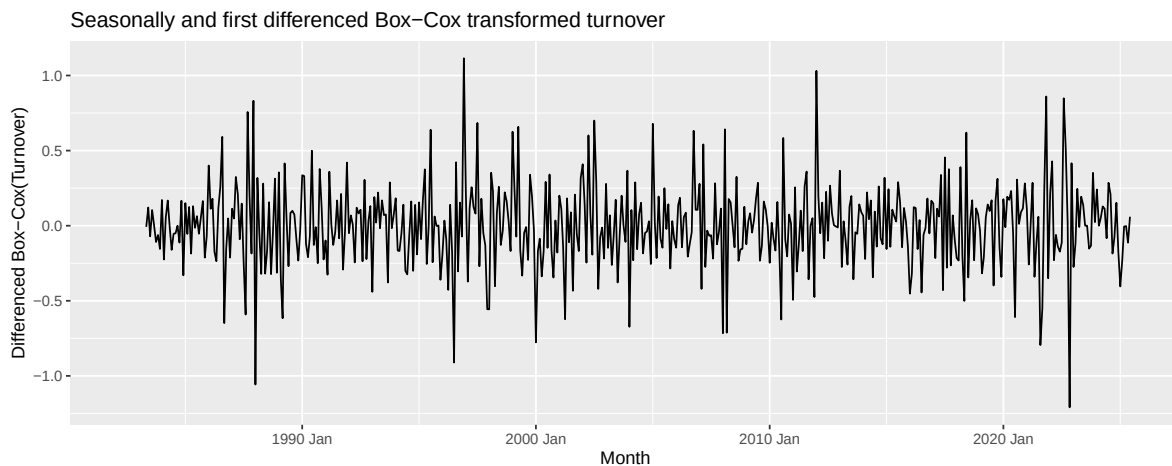


Figure 6: Seasonally and first differenced Box-Cox transformed turnover series.

Figure 6 shows that after applying both seasonal and ordinary differencing, the series fluctuates more closely around zero with no clear remaining trend or seasonal pattern. Although some temporary shocks remain, this looks more stationary than using seasonal differencing alone. I therefore use one seasonal difference and one first difference, so $D = 1$, $d = 1$, and $m = 12$.

```
seasonal_diff_tbl <- print_retail_act |>
  features(box_cox(y, lambda), unitroot_nsdiffs)

first_diff_tbl <- print_retail_act |>
  features(difference(box_cox(y, lambda), 12), unitroot_ndiffs)

bind_cols(seasonal_diff_tbl, first_diff_tbl) |>
  kable()
```

Table 2: Unit root checks used as supporting evidence for differencing.

nsdifs	ndiffs
1	1

Table 2 also supports this choice, suggesting one seasonal difference and one ordinary difference.

```
kpss_seasonal <- print_retail_act |>
  features(difference(box_cox(y, lambda), 12), unitroot_kpss) |>
  mutate(series = "After seasonal difference only")

kpss_final <- print_retail_act |>
  features(difference(box_cox(y, lambda), 12) |> difference(), unitroot_kpss) |>
  mutate(series = "After seasonal and first difference")

bind_rows(kpss_seasonal, kpss_final) |>
  select(series, kpss_stat, kpss_pvalue) |>
  mutate(across(where(is.numeric), ~ round(.x, 3))) |>
  kable()
```

Table 3: KPSS tests for the transformed series after differencing.

series	kpss_stat	kpss_pvalue
After seasonal difference only	0.645	0.019
After seasonal and first difference	0.011	0.100

Table 3: KPSS tests for the transformed series after differencing.

series	kpss_stat	kpss_pvalue
--------	-----------	-------------

Table 3 provides an additional stationarity check. Since the KPSS null hypothesis is stationarity, the seasonally differenced series is still not stationary at the 5% level ($p = 0.019$). After both seasonal and first differencing, I fail to reject stationarity ($p = 0.100$), supporting $D = 1$, $d = 1$, and $m = 12$.

ARIMA model selection and diagnostics

```
print_retail_act |>
  gg_tsdisplay(difference(box_cox(y, lambda), 12) |> difference(),
               plot_type = "partial",
               lag_max = 60) +
  labs(title = "Stationary transformed series after seasonal and first differencing")
```

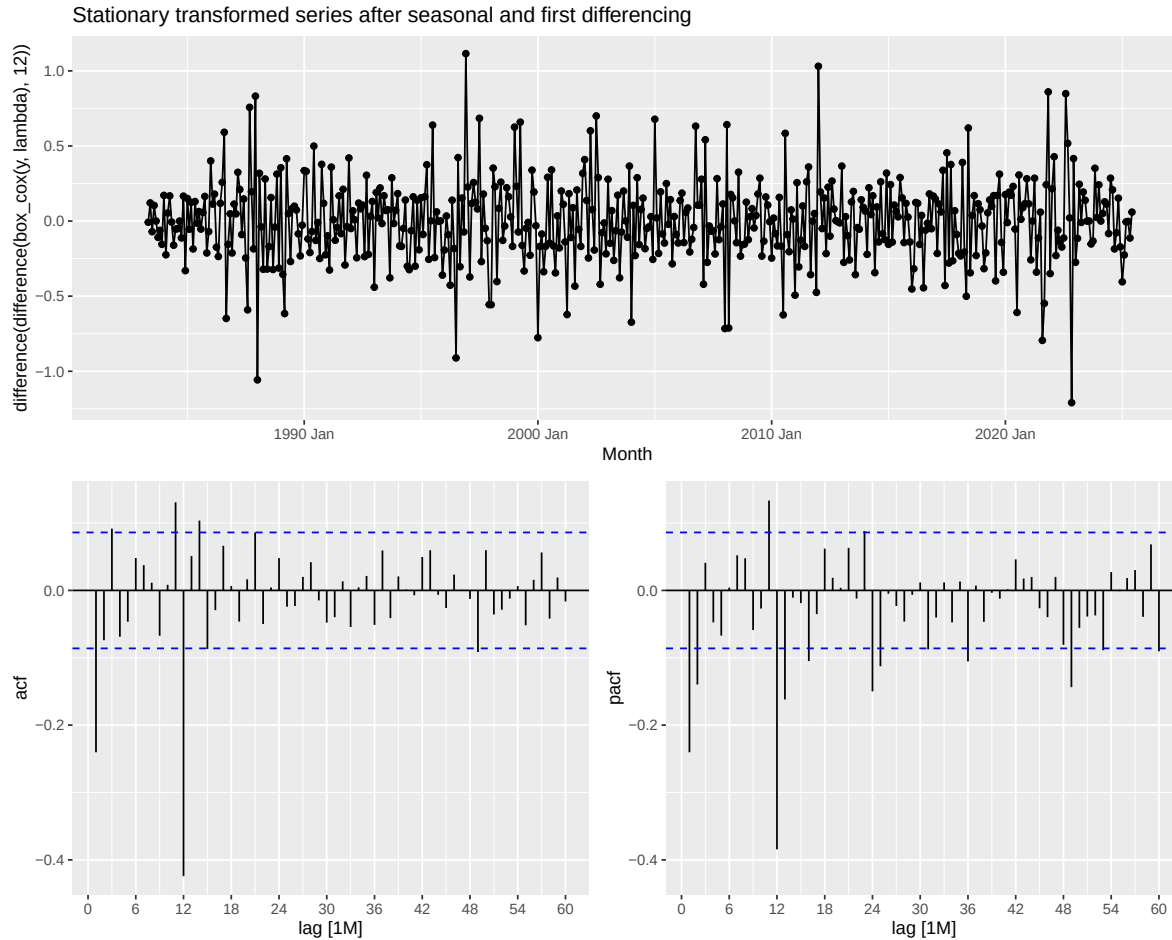


Figure 7: ACF and PACF of the stationary transformed series.

Figure 7 shows the ACF and PACF after the Box-Cox transformation, seasonal differencing and first differencing. The ACF has a strong spike at lag 1 and at the seasonal lag 12, suggesting non-seasonal MA(1) and seasonal MA(1). The PACF also shows early spikes up to about lag 2, so AR(2) is a plausible alternative. I therefore use $ARIMA(0,1,1)(0,1,1)_{12}$ as the initial model and compare it with plausible alternatives using AICc.

```
q2_arma_candidates <- print_retail_act |>
  model(
    arima011011 = ARIMA(box_cox(y, lambda) ~ pdq(0,1,1) + PDQ(0,1,1)),
    arima210011 = ARIMA(box_cox(y, lambda) ~ pdq(2,1,0) + PDQ(0,1,1)),
    arima011110 = ARIMA(box_cox(y, lambda) ~ pdq(0,1,1) + PDQ(1,1,0)),
    arima011210 = ARIMA(box_cox(y, lambda) ~ pdq(0,1,1) + PDQ(2,1,0)),
    arima011310 = ARIMA(box_cox(y, lambda) ~ pdq(0,1,1) + PDQ(3,1,0)),
```

```

    arima210110 = ARIMA(box_cox(y, lambda) ~ pdq(2,1,0) + PDQ(1,1,0)),
    arima210210 = ARIMA(box_cox(y, lambda) ~ pdq(2,1,0) + PDQ(2,1,0)),
    arima210310 = ARIMA(box_cox(y, lambda) ~ pdq(2,1,0) + PDQ(3,1,0))
  )

```

```

q2_arima_candidates |>
  glance() |>
  arrange(AICc) |>
  select(.model, sigma2, log_lik, AIC, AICc, BIC) |>
  mutate(across(where(is.numeric), ~ round(.x, 3))) |>
  kable()

```

Table 4: Information criteria for candidate ARIMA models.

.model	sigma2	log_lik	AIC	AICc	BIC
arima011011	0.055	13.956	-21.911	-21.863	-9.231
arima210011	0.055	14.542	-21.084	-21.004	-4.178
arima0111310	0.057	8.008	-6.016	-5.896	15.117
arima210310	0.057	8.719	-5.439	-5.271	19.920
arima011210	0.057	4.483	-0.965	-0.885	15.941
arima210210	0.058	4.889	0.221	0.341	21.354
arima0111110	0.059	-3.302	12.603	12.651	25.283
arima210110	0.059	-2.966	13.932	14.012	30.839

Based on Table 4, ARIMA(0,1,1)(0,1,1)₁₂ has the lowest AICc, so it is the best candidate model among those considered. The next closest model is ARIMA(2,1,0)(0,1,1)₁₂, but its AICc is slightly higher. Both best-performing models include seasonal MA(1), while the models using seasonal AR terms have much higher AICc values. This suggests that the seasonal MA(1) structure is more appropriate than the seasonal AR alternatives.

I therefore treat ARIMA(0,1,1)(0,1,1)₁₂ as the best manual candidate model and check its residual diagnostics before making the final model decision.

```

q2_manual_arima <- q2_arima_candidates |>
  select(arima011011)

q2_manual_arima |>
  report()

```

Series: y

```
Model: ARIMA(0,1,1)(0,1,1)[12]
Transformation: box_cox(y, lambda)
```

```
Coefficients:
```

```
          ma1      sma1
      -0.3249 -0.6100
s.e.    0.0463  0.0452
```

```
sigma^2 estimated as 0.055:  log likelihood=13.96
AIC=-21.91  AICc=-21.86  BIC=-9.23
```

The estimated MA coefficient is -0.3249 and the seasonal MA coefficient is -0.6100. Both are negative, suggesting that recent non-seasonal and seasonal shocks are partly corrected in the following periods. This model is simple and is also consistent with the ACF pattern in Figure 7.

```
q2_manual_arima |>
  gg_tsresiduals(lag = 48)
```

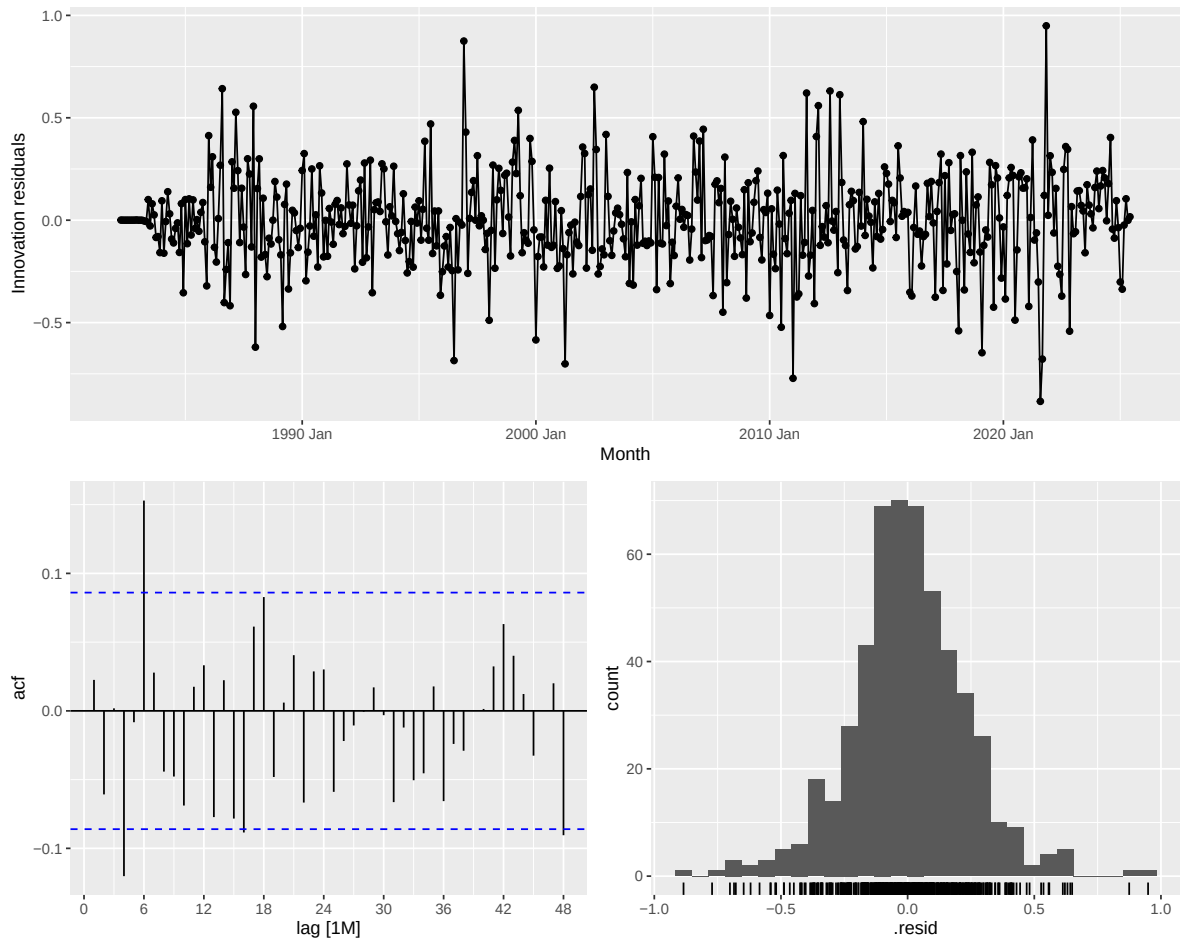


Figure 8: Residual diagnostics for the manual $\text{ARIMA}(0,1,1)(0,1,1)[12]$ candidate model.

Figure 8 shows that the residual time plot has no obvious remaining trend or seasonal pattern. The histogram is roughly symmetric and reasonably bell-shaped, although not perfectly normal. The residual ACF still has some significant spikes, so the residuals are not completely clean visually. Therefore, I use the Ljung–Box test as a formal check for remaining autocorrelation.

```
q2_manual_arima |>
  augment() |>
  features(.innov, ljung_box, lag = 24, dof = 2) |>
  kable(digits = 3)
```

Table 5: Ljung-Box test for residual whiteness of the manual ARIMA candidate model.

.model	lb_stat	lb_pvalue
arima011011	50.268	0.001

For Table 5, I use lag 24 and dof 2, from one non-seasonal MA term and one seasonal MA term. The null hypothesis is $H_0 : \rho_1 = \dots = \rho_{24} = 0$ jointly, meaning the residual autocorrelations up to lag 24 are jointly insignificant. Since $p = 0.001 < 0.05$, I reject H_0 at the 5% level. This suggests that the residuals are not white noise and that some autocorrelation remains. Therefore, I review the model rather than using this candidate model directly as the final forecasting model.

```
q2_auto_arima <- print_retail_act |>
  model(auto_arima = ARIMA(box_cox(y, lambda),
                           stepwise = FALSE,
                           approximation = FALSE))

q2_auto_arima |>
  report()
```

```
Series: y
Model: ARIMA(3,1,1)(1,1,1)[12]
Transformation: box_cox(y, lambda)
```

```
Coefficients:
      ar1      ar2      ar3      ma1      sar1      sma1
    0.6688  0.1202  0.1180 -0.9845  0.2179 -0.7672
s.e.  0.0454  0.0533  0.0448  0.0109  0.0896  0.0723
```

```
sigma^2 estimated as 0.05353:  log likelihood=21.47
AIC=-28.94  AICc=-28.71  BIC=0.65
```

The manual candidate model failed the Ljung-Box test, so I use a full automatic ARIMA search with `stepwise = FALSE` and `approximation = FALSE`. The automatic search selects $\text{ARIMA}(3,1,1)(1,1,1)_{12}$ with the same Box-Cox transformation. It also uses $d = 1$ and $D = 1$, which agrees with the earlier differencing decision.

```
q2_compare_arima <- print_retail_act |>
  model(manual_arima = ARIMA(box_cox(y, lambda) ~ pdq(0,1,1) + PDQ(0,1,1)),
        auto_arima = ARIMA(box_cox(y, lambda),
```



```

      stepwise = FALSE,
      approximation = FALSE))

q2_compare_arima |>
  glance() |>
  arrange(AICc) |>
  select(.model, sigma2, log_lik, AIC, AICc, BIC) |>
  mutate(across(where(is.numeric), ~ round(.x, 3))) |>
  kable()

```

Table 6: Information criteria comparison between the manual candidate ARIMA model and the automatic ARIMA model.

.model	sigma2	log_lik	AIC	AICc	BIC
auto_arima	0.054	21.470	-28.939	-28.715	0.646
manual_arima	0.055	13.956	-21.911	-21.863	-9.231

Table 6 shows that the automatic ARIMA model has a lower AICc than the manual candidate model. Since both models use the same Box-Cox transformation and differencing orders, the AICc comparison is appropriate. The automatic model therefore appears better, but I still check its residual diagnostics before using it for forecasting.

```

q2_auto_arima |>
  gg_tsresiduals(lag = 48)

```

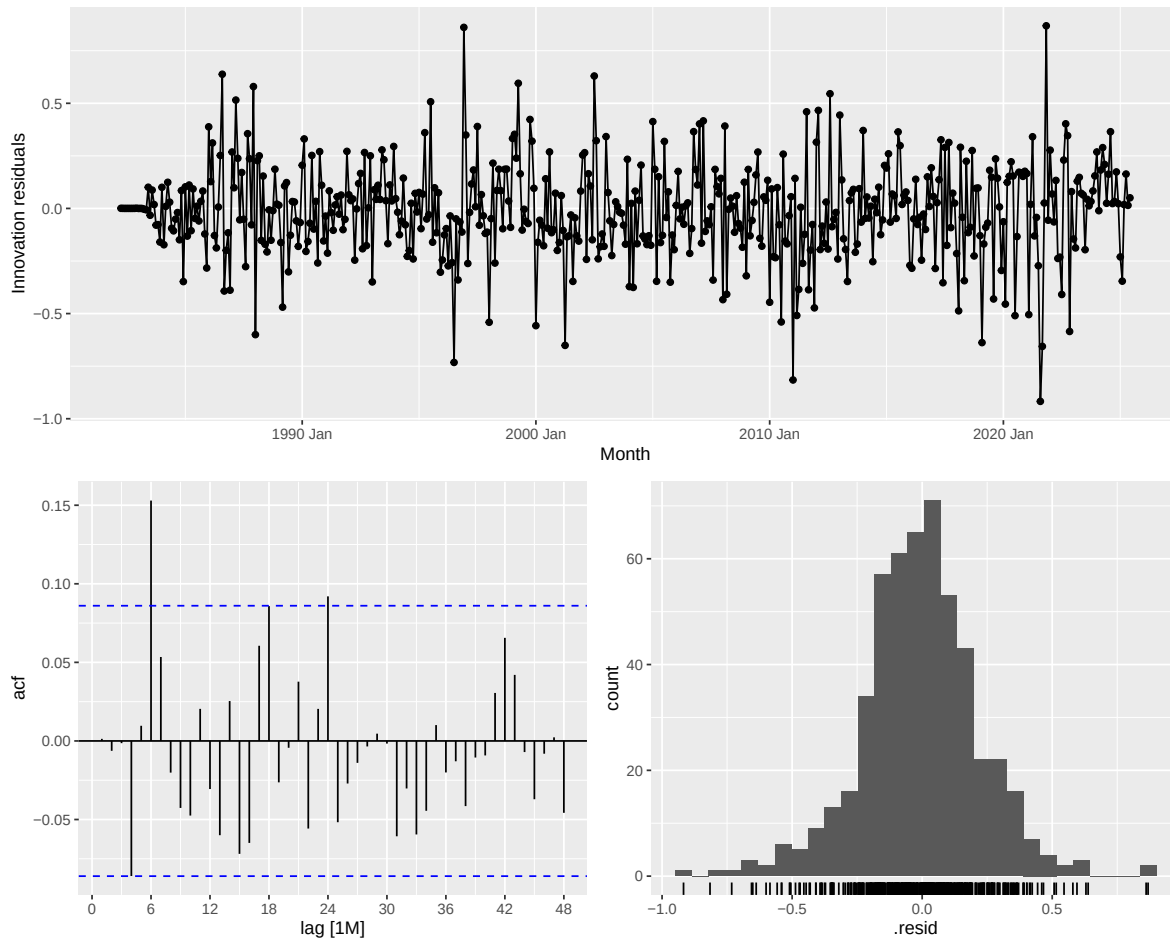


Figure 9: Residual diagnostics for the automatic $\text{ARIMA}(3,1,1)(1,1,1)_{12}$ model.

Figure 9 shows the residual diagnostics for the automatic $\text{ARIMA}(3,1,1)(1,1,1)_{12}$ model. The residual time plot has no obvious trend or seasonal pattern, and the histogram is roughly symmetric and reasonably bell-shaped. However, the residual ACF still has a few significant spikes, so I use the Ljung–Box test as a formal check for remaining autocorrelation.

```
q2_auto_arima |>
  augment() |>
  features(.innov, ljung_box, lag = 24, dof = 6) |>
  kable(digits = 3)
```

Table 7: Ljung-Box test for residual whiteness of the automatic ARIMA model.

.model	lb_stat	lb_pvalue
auto_arima	41.841	0.001

For Table 7, I use lag 24 and dof 6, from three non-seasonal AR terms, one non-seasonal MA term, one seasonal AR term and one seasonal MA term. The null hypothesis is $H_0 : \rho_1 = \dots = \rho_{24} = 0$ jointly. Since $p = 0.001 < 0.05$, I reject H_0 at the 5% level. This suggests that some residual autocorrelation remains, so the automatic model is not perfect.

Final Decision

The manual model is simpler, but I select the automatic ARIMA(3,1,1)(1,1,1)₁₂ model because it has the lower AICc. This is important because AICc already penalises the extra parameters, so the lower AICc suggests that the additional terms improve the fit. However, the Ljung-Box test still rejects white noise, so the model is not perfect and the forecasts should be interpreted cautiously.

Forecasts from the final ARIMA model

```
q2_final_arima <- q2_auto_arima

q2_arima_fc <- q2_final_arima |>
  forecast(h = "24 months")

forecast_start <- max(print_retail_act$Month) + 1

q2_arima_fc |>
  autoplot(print_retail_act, level = c(80, 95)) +
  labs(title = "Twenty-four month ARIMA forecast",
       subtitle = "Final model: ARIMA(3,1,1)(1,1,1)[12] with Box-Cox transformation",
       x = "Month",
       y = "Turnover (Millions of Dollars)")
```

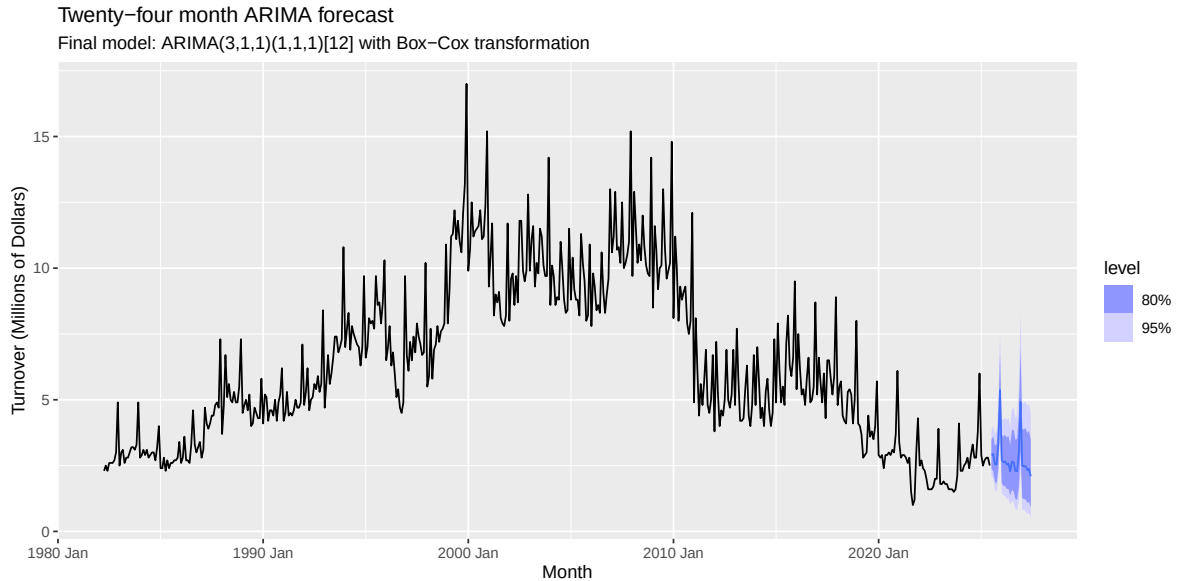


Figure 10: Twenty-four month forecasts from the final ARIMA model over the full updated series.

Figure 10 shows the 24-month forecasts from the final $\text{ARIMA}(3,1,1)(1,1,1)_{12}$ model in the context of the full series. The forecasts look reasonable because they continue the recent lower-turnover period, which is consistent with the long-run downward movement seen in the later part of the series.

```
q2_arima_fc |>
  autoplot(print_retail_act |> filter(year(Month) >= 2018),
    level = c(80, 95)) +
  geom_vline(xintercept = forecast_start,
    linetype = "dashed") +
  labs(title = "Zoomed twenty-four month ARIMA forecast",
    subtitle = "Dashed line marks the start of the forecast period",
    x = "Month",
    y = "Turnover (Millions of Dollars)")
```

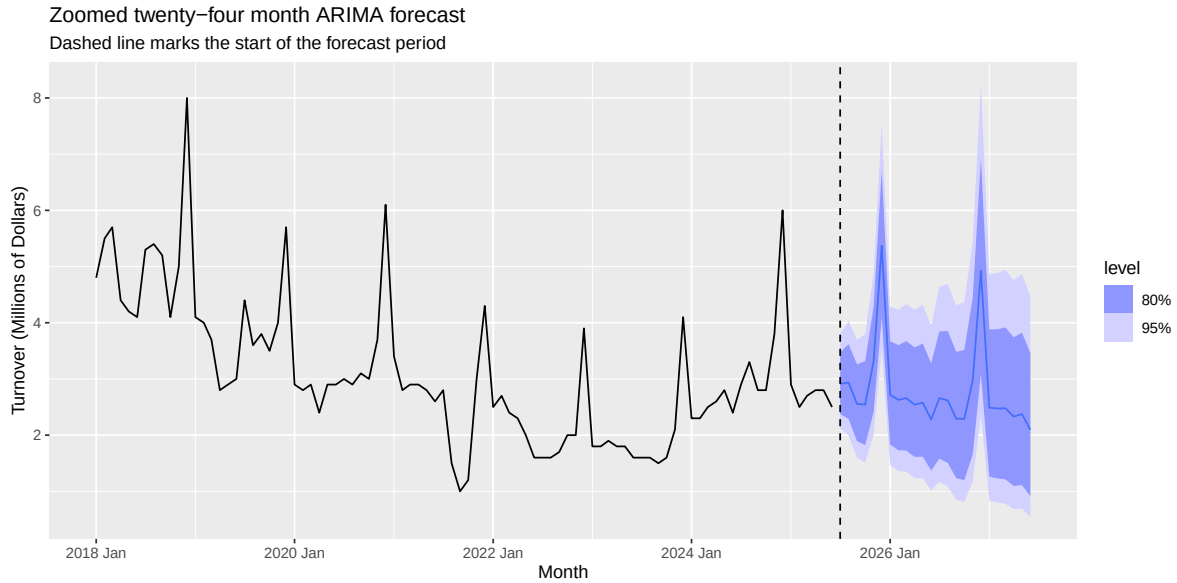


Figure 11: Zoomed twenty-four month forecasts from the final ARIMA model.

Figure 11 gives a clearer view of the recent data and forecast horizon. The point forecasts continue from the latest observed level and retain strong monthly seasonality, with expected seasonal peaks and troughs.

The prediction intervals widen as the forecast horizon increases, reflecting greater uncertainty further into the future. Since the final ARIMA model still failed the Ljung–Box test, the forecasts should be interpreted cautiously. The model is the best ARIMA choice among those considered based on AICc, but the remaining residual autocorrelation suggests that the forecast intervals may not fully capture all uncertainty.

Question 3

On the same graph, plot the updated data together with forecasts from all the ‘best’ alternative methods/models you have studied throughout the semester. Generate as many steps ahead forecasts required to cover the new data. Think about visualisation. Visually inspect these and comment on any similarities, differences, anomalies you may observe. (No more than 100 words). (12 marks)

```
# Training set = data available at the time of GA2/GA3 (up to 2023 Jun)
train_data <- print_retail_act |>
  filter(Month <= yearmonth("2023 Jun"))

# Define the transformation
lambda_train <- train_data |>
  features(y, guerrero) |>
  pull(lambda_guerrero)

# Refit the models
fit_all <- train_data |>
  model(
    `Benchmark (SNAIVE)` = SNAIVE(y),
    `Regression (TSLM)` = TSLM(box_cox(y, lambda_train) ~ trend() + season()),
    `STL Decomposition` = decomposition_model(
      STL(box_cox(y, lambda_train) ~ season(window = "periodic")),
      NAIVE(season_adjust)),
    `Best ETS (M,N,M)` = ETS(y ~ error("M") + trend("N") + season("M")),
    `Best ARIMA` = ARIMA(box_cox(y, lambda_train) ~ pdq(3,1,1) + PDQ(1,1,1))

fc_all <- fit_all |>
  forecast(h = "24 months")
```

```
fc_all |>
  autoplot(print_retail_act |> filter_index("2021 Jan" ~ .),
    level = NULL, linewidth = 0.8) +
  geom_vline(xintercept = yearmonth("2023 Jul"),
    linetype = "dashed",
    colour = "black",
    linewidth = 0.8) +
  labs(title = "Forecast comparison using models fitted to June 2023",
    subtitle = "Dashed line marks the start of the updated data period",
    y = "Turnover (Millions of Dollars)", x = "Month", colour = "Model") +
  theme_minimal() +
  theme(legend.position = "bottom")
```

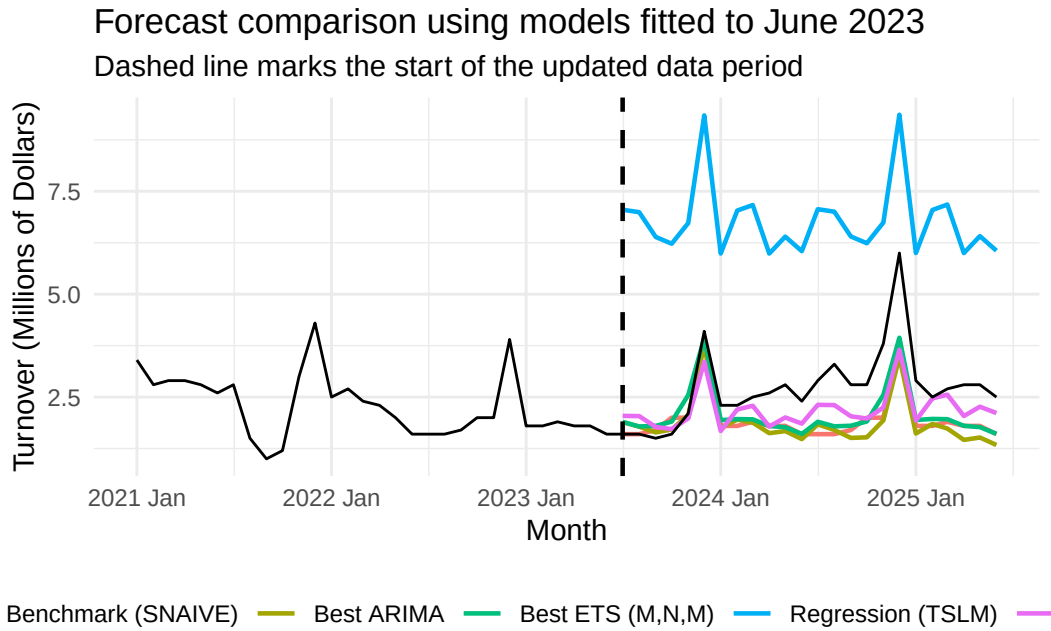


Figure 12: Updated turnover series to June 2025 with 24-month-ahead forecasts from selected models fitted on data up to June 2023. The dashed vertical line marks the start of the updated data period.

SNAIVE, TSLM, STL, ETS and ARIMA were fitted only to data up to June 2023 before forecasting the updated period. Figure 12 shows that all models retain seasonal movement but differ in levels and peak sizes. TSLM systematically over-forecasts, while STL tracks non-peak months most closely. ETS and SNAIVE capture the December peak timing well but still under-forecast magnitudes. ARIMA performs reasonably initially but under-forecasts more clearly by late 2024. The sharp December 2024 peak is the main anomaly, as none of the models forecast it well.

Question 4

Summarise the accuracy of the competing forecasts over the new/updated data and choose the best two models/methods from this evaluation. (6 marks)

```
accuracy_tbl <- fc_all |>
  accuracy(print_retail_act) |>
  filter(.type == "Test") |>
  arrange(RMSE)

accuracy_tbl |>
  mutate(across(where(is.numeric), ~ round(.x, 3))) |>
  kable()
```

Table 8: Forecast accuracy over the updated data period, July 2023 to June 2025.

.model	.type	ME	RMSE	MAE	MPE	MAPE	MASE	RMSSEACF1	
STL	Test	0.524	0.806	0.630	14.741	21.490	0.598	0.542	0.621
Decomposition									
Best ETS	Test	0.639	0.886	0.767	19.599	27.154	0.729	0.596	0.751
(M,N,M)									
Benchmark	Test	0.758	0.971	0.808	24.507	27.701	0.768	0.653	0.755
(SNAIVE)									
Best ARIMA	Test	0.854	1.095	0.915	27.420	31.270	0.869	0.737	0.766
Regression	Test	-4.068	4.136	4.068	-	170.212	3.864	2.782	0.510
(TSLM)					170.212				

The forecasts produced from the models fitted in Q3 are evaluated against the updated observations from July 2023 to June 2025. Based on the accuracy results in Table 8, the STL decomposition model is the best-performing model, with the lowest RMSE, MAE, MAPE, MASE and RMSSE. ETS(M,N,M) is the second-best model across the same measures. SNAIVE and ARIMA have larger forecast errors, while TSLM performs worst, suggesting that a simple trend and seasonal regression was not flexible enough for the updated period. Therefore, STL decomposition and ETS(M,N,M) are selected for the final forecasts.

Question 5

Use the best two models and generate forecasts for your series based on the end of your updated sample. Plot these (think about visualisation). Comment on these. (No more than 100 words). (6 marks)

```
fit_best_two <- print_retail_act |>
  model(
    `STL Decomposition` = decomposition_model(
      STL(box_cox(y, lambda) ~ season(window = "periodic")),
      NAIVE(season_adjust)
    ),
    `ETS(M,N,M)` = ETS(y ~ error("M") + trend("N") + season("M"))
  )

fc_best_two <- fit_best_two |>
  forecast(h = "24 months")

fc_best_two |>
  autoplot(
    print_retail_act |> filter(Month >= yearmonth("2018 Jan")),
    level = c(80, 95),
    alpha = 0.5
  ) +
  labs(
    title = "Forecasts from the two best-performing models",
    subtitle = "STL decomposition and ETS(M,N,M), refitted using the full updated sample",
    x = "Month",
    y = "Turnover (Millions of Dollars)",
    colour = "Model",
    fill = "Model"
  ) +
  theme_minimal() +
  theme(legend.position = "bottom")
```

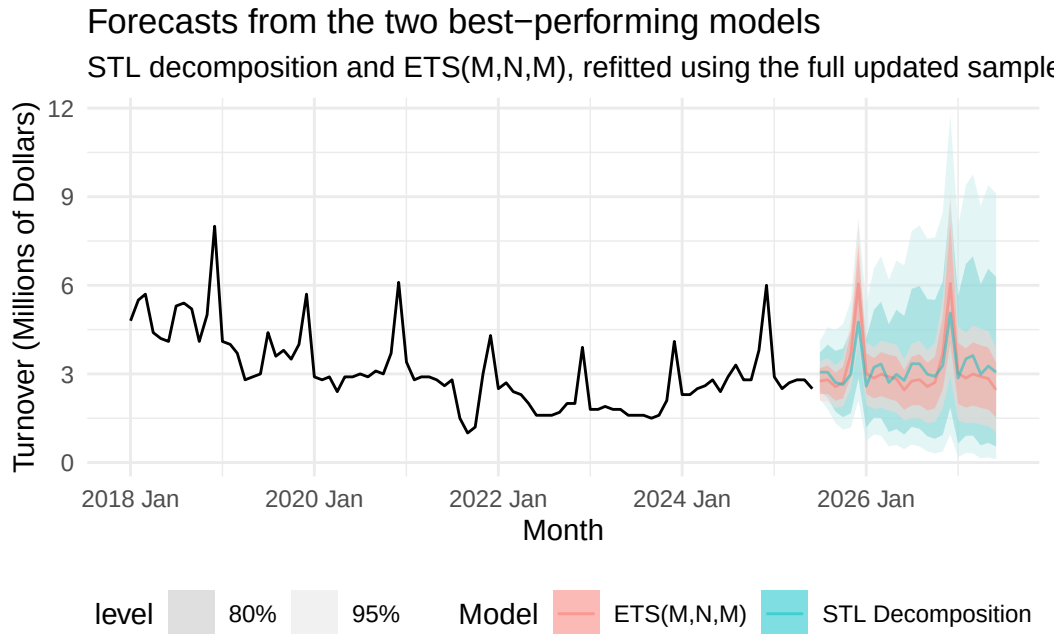


Figure 13: Forecasts from the two best-performing models fitted to the full updated sample.

Using the two best models selected in Q4, STL decomposition and ETS(M,N,M) were refitted to the full updated sample before producing 24-month forecasts. In Figure 13, both models forecast turnover around the recent post-COVID level and retain strong end-of-year seasonal peaks. STL shows stronger seasonal fluctuations and wider prediction intervals, suggesting greater uncertainty from recent volatility. ETS(M,N,M) has narrower intervals and a smoother overall path, while still producing clear end-of-year peaks. Therefore, future seasonal peaks should be interpreted cautiously.

Question 6

You have now applied a range of modelling frameworks throughout the semester and developed forecasts for multiple time series. For this final task, write a short report (300–500 words) reflecting on your thoughts and experiences with these frameworks. Your report should address aspects such as: which methods worked best for the series you analysed; which framework did you find the easiest to understand and implement; what were the key strengths and limitations of each approach. You should consider including at least one comment on each of: benchmarks, decomposition, ETS, ARIMA and regression frameworks. Additionally, reflect on whether a regression-based model might be more suitable for forecasting retail data affected by COVID-19, and explain how you would approach this implementation. (20 marks)

This task is intentionally open-ended to give you the opportunity to think critically and share your personal insights. Go beyond the questions above if you can, consider what you enjoyed, what you didn't, what challenged you, and what you've learnt from all your hard work. Treat this as a report advising/informing a client (who knows statistics) while you were working for your forecasting consulting group. Include your consulting group name.

Congratulations on completing all tasks for this unit. I hope you've enjoyed the learning journey and gained valuable forecasting skills. Well done on all your hard work!

Cheers, George and team.

As GenieUs group, our forecasting work showed that the best model depends on the structure of the time series rather than on model complexity alone. A suitable forecasting method should match the main features of the historical data, including level, trend, seasonality, volatility and unusual shocks. For our retail series, the accuracy results in Table 8 showed that STL decomposition and ETS(M,N,M) performed best. These models captured the recent level and recurring seasonal pattern more effectively than the benchmark, ARIMA and regression models.

Benchmark methods, such as seasonal naïve, were useful because they provided a simple reference point. They are easy to understand and implement, and they help check whether more advanced models actually improve forecast accuracy. However, benchmarks are limited because they mainly repeat past patterns and cannot adapt well to structural changes.

Decomposition was one of the easiest frameworks to interpret. STL was helpful because it separated the series into trend, seasonal and remainder components, making it easier to explain the pattern of the data to a client. In our case, STL worked well because the series had strong seasonality and a changing recent level. Its limitation is that the forecasts can show wider uncertainty when the historical series is volatile.

ETS was also intuitive because it models the level, trend and seasonality directly. ETS(M,N,M) performed strongly for our series because the recent data showed strong multiplicative seasonality but no clear long-term trend. ETS is useful for automatic forecasting and generally produces

smooth forecasts, but selecting between additive and multiplicative components still requires careful visual judgement.

ARIMA was the most challenging framework because it requires understanding stationarity, differencing, ACF/PACF patterns and residual diagnostics. Its strength is flexibility, as it can capture many autocorrelation patterns. However, ARIMA is less visually intuitive than decomposition or ETS, and for our series it did not forecast as accurately as the best two models.

Regression models were useful because they allow external information to be included. A simple TSLM model was less suitable for our retail series because fixed trend and seasonal terms could not fully adapt to the structural change and volatility after COVID-19. However, a regression-based model may be more suitable for retail data affected by COVID-19 if relevant predictors are included. We would approach this using dynamic regression with ARIMA errors, adding intervention variables such as lockdown periods, reopening periods, online-shopping effects, inflation or consumer spending indicators. The regression part would capture COVID-related impacts, while ARIMA errors would handle remaining autocorrelation.

Overall, we learnt that forecasting is not about choosing the most complex model. The best model is the one whose assumptions fit the data and performs well through visual checks, residual diagnostics and out-of-sample accuracy.